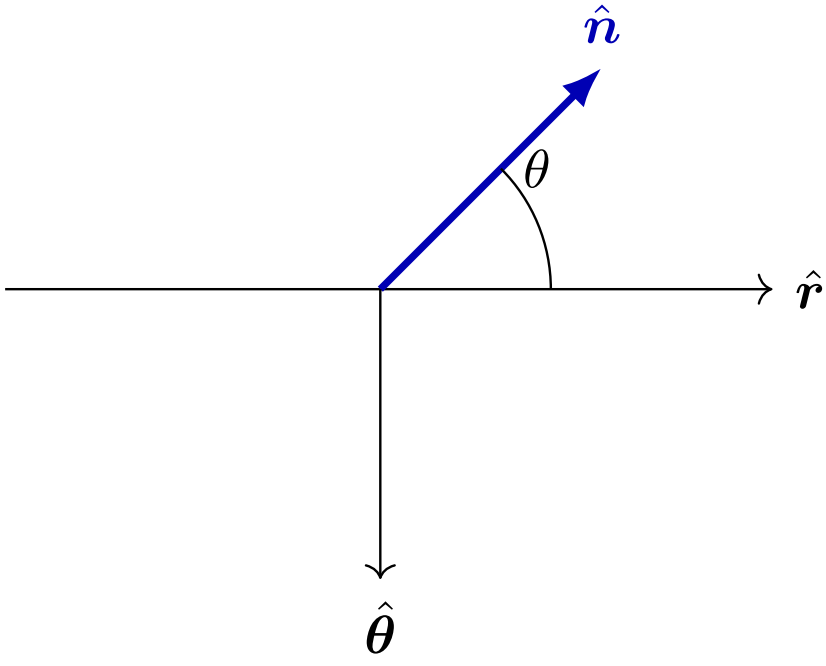
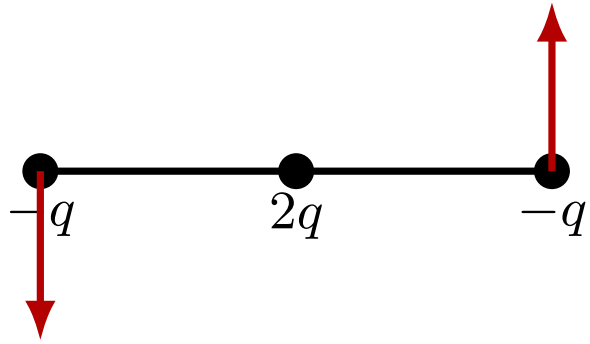


$$U = K(1 - 3\cos^2\theta)/r^3$$

$$F_r = 3K(1 - 3\cos^2\theta)/r^4$$

$$F_\theta = -6K\sin\theta\cos\theta/r^4$$

$\odot\, \boldsymbol{b}$（垂直分子）



两端力相反、力矩同向

$$\boldsymbol{\tau} = ql^2[\boldsymbol{b} - \hat{\boldsymbol{n}}(\hat{\boldsymbol{n}} \cdot \boldsymbol{b})]$$

$\boldsymbol{b} \parallel \hat{\boldsymbol{n}}$ 时各端电场为零